A degree-18 counterexample to the group-order bound for phylogenetic complexity

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Abstract

We exhibit two compatible 18 x 3 tables of zero-sum rows over $G = \mathbb{Z}/17\mathbb{Z}$. Their column multisets admit exactly these two tables, up to row permutation, and the tables have disjoint row supports. Consequently, no sequence of moves involving at most 17 rows connects them. Thus $\phi(\mathbb{Z}/17\mathbb{Z}, 3) \geq 18$, contradicting the proposed bound $\phi(G) \leq |G|$. The proof consists of explicit data and elementary rational linear algebra. We also give a direct ideal-theoretic proof, a boundary-point obstruction even to set-theoretic generation in degrees at most 17, and reproducible exact checks.

1 Statement and a general obstruction

A flow of length n over a finite abelian group G is a tuple $(g_1, \ldots, g_n)$ with $\sum_j g_j = 0$ in G. Two tables of flows are compatible if their corresponding columns are the same multisets. A move of size at most d replaces at most d rows with a compatible collection of flows. Tables are considered up to permutation of their rows.

This is the table formulation of ideal generation in [1, Section 2]. The proposed group-order bound and its three-leaf special case appear as Conjectures 4.1 and 4.2 there; see also [2, Conjecture 27 in the arXiv version].

Theorem 1. The tables A, B in Table 1 are compatible tables over $\mathbb{Z}/17\mathbb{Z}$, each with 18 rows and three columns. Their common column multisets admit exactly two tables, namely A and B, up to row permutation. Every move changing either table has size 18. In particular,

$$\phi(\mathbb{Z}/17\mathbb{Z}, n) \geq 18 \qquad \text{for every } n \geq 3, \qquad \phi(\mathbb{Z}/17\mathbb{Z}) > 17.$$

Lemma 2. Suppose a collection of fixed column multisets admits exactly two m-row tables of flows A, B. If A and B have no row type in common, they cannot be connected by moves of size less than m.

Proof. A move of size less than m preserves at least one original row, so cannot take A to B. Every move preserves the column multisets; hence its result is again A or B. No first nontrivial move from A is possible, which excludes a chain of any length.

3 An elementary kernel computation

Let M be the 23×22 incidence matrix whose rows are labelled by (j, v), with $j = 1, 2, 3$ and $v \in S_j$ in increasing order, and whose columns are the types r_i:

$$M_{(j,v),i} = \begin{cases} 1, & (r_i)_j = v, \\ 0, & \text{otherwise.} \end{cases}$$

Thus Mx records the column multiplicities of a table with type counts x. All equalities involving M below are over $\mathbb{Q}$, not modulo 17.

Lemma 3. $\ker_{\mathbb{Q}} M = \mathbb{Q}(\beta - \alpha)$.

Proof. Let $z \in \ker_{\mathbb{Q}} M$ and put

$$a = z_1, \quad b = z_2, \quad d = z_6, \quad e = z_7, \quad f = z_9, \quad g = z_{11}, \quad h = z_{17}.$$

The marginal equations for the first two columns give, by successive substitution,

$$\begin{aligned} z = (&a, b, -a - b, -a, a - d, d, e, -e, f, b - f - g, g, -b, \\ &g + d, -g, -d, -h - a - b - e, h, a + b + e, -f, \\ &-g - d + h + a + b + e, -b + f + g - h, -a + d - e). \end{aligned}$$

Conversely, this expression satisfies all first- and second-column equations for arbitrary values of the seven parameters. The third-column equations for the values 14, 7, 11, 2, 5, 13, in that order, are

$$g + h = 0, \qquad -a - d + f = 0, \qquad -a + 2d - 2e = 0,$$
$$-b - d - e - f - h = 0, \qquad a + 2b + d + e + g = 0, \qquad a - 2b + f - h = 0.$$

The first three give $h = -g$, $f = a + d$, and $a = 2d - 2e$. The fourth then gives $g = b + 4d - e$, while the last two give

$$7d - 2e + 3b = 0, \qquad b = 9d - 5e.$$

Consequently,

$$0 = 7d - 2e + 3(9d - 5e) = 34d - 17e,$$

so $e = 2d$. Substitution gives

$$(a, b, d, e, f, g, h) = d(-2, -1, 1, 2, -1, 1, -1),$$

and hence

$$\begin{aligned} z = d(&-2, -1, 3, 2, -3, 1, 2, -2, -1, -1, 1, 1, \\ &2, -1, -1, 2, -1, -1, 1, -4, 2, 1) = d(\beta - \alpha). \end{aligned}$$

The remaining third-column equation, for value 1, is $b - d + 2e - f - 2g + h = 0$ and is satisfied. Since the compatible vectors α, β are distinct, $\beta - \alpha$ is a nonzero element of the kernel. This proves the assertion.

Proof of Theorem 1. Any compatible table has a count vector $x \in \mathbb{Z}^{22}_{\geq 0}$ with $Mx = M\alpha$. By Lemma 3,

$$x = \alpha + t(\beta - \alpha), \qquad t \in \mathbb{Q}.$$

The sixth coordinate gives $x_6 = t \in \mathbb{Z}_{\geq 0}$, and the second gives $x_2 = 1 - t \geq 0$. Thus $t \in \{0, 1\}$ and x is α or β. Lemma 2 excludes every move of size at most 17. Replacing all 18 rows does work. Appending zero columns preserves exactly the same compatibility problem, proving the assertion for every $n \geq 3$.

4 A direct ideal-theoretic certificate

This section independently explains why the obstruction concerns the full group-based ideal, not merely a chosen subset of moves. Let k be a field and put

$$F = \{(a, b, c) \in (\mathbb{Z}/17\mathbb{Z})^3 : a + b + c = 0\}, \qquad R = k[X_r : r \in F].$$

The three-leaf group-based toric ideal is the kernel I of the map

$$X_{(a,b,c)} \mapsto U^a V^b W^c .$$

Let $E = \{r_1, \ldots, r_{22}\}$ be the allowed types from Table 1, write $X_i = X_{r_i}$, and set $R_E = k[X_1, \ldots, X_{22}]$. Consider

$$F = X^{\alpha} - X^{\beta} = X_1^2 X_2 X_5^3 X_8^2 X_9 X_{10} X_{14} X_{15} X_{17} X_{18} X_{20}^4 - X_3^3 X_4^2 X_6 X_7^2 X_{11} X_{12}^2 X_{13}^2 X_{16} X_{19} X_{21}^2 X_{22} .$$

Both monomials have degree 18, and $F \in I$ by compatibility.

Proposition 4. For the specialization $\pi : R \to R_E$ which sets all variables outside E to zero, one has

$$\pi(I) = I \cap R_E = (F).$$

In particular, every element of I of total degree at most 17 specializes to zero, whereas $\pi(F) = F \neq 0$.

Proof. First consider a binomial $X^u - X^v \in I$. If one monomial uses only variables in E, it has no column label outside the sets S_j. Its equal column multiplicities force the other monomial to have this same property. As E contains all flows with labels in S_j, the other monomial also uses only E. Thus π either retains both monomials or kills both. The kernel of a monomial map is spanned by binomials with equal images: this follows simply by grouping terms with the same image monomial. It follows that $\pi(I) = I \cap R_E$.

For any binomial in this restricted ideal, its exponent difference lies in $\ker_{\mathbb{Z}} M$. Lemma 3, and the coordinate $(\beta - \alpha)_6 = 1$, give

$$\ker_{\mathbb{Z}} M = \mathbb{Z}(\beta - \alpha).$$

After interchanging the two monomials if necessary and factoring their greatest common monomial divisor, the binomial is therefore a monomial multiple of

$$(X^{\alpha})^m - (X^{\beta})^m, \qquad m \geq 1.$$

This polynomial is divisible by F. Conversely, F belongs to the restricted ideal. Hence the restricted ideal is (F). A nonzero polynomial multiple of the homogeneous degree-18 polynomial F cannot have total degree less than 18, which proves the last claim.

Were I generated by polynomials of degrees at most 17, their specializations would generate $\pi(I)$. All would be zero, although $\pi(I) = (F) \neq 0$. This is a direct contradiction. It establishes the same degree lower bound without invoking a connectivity theorem.

5 A stronger boundary obstruction and exact checks

Corollary 5. Let $J_{\le 17} \subset I$ be the ideal generated by all elements of I of total degree at most 17. Then

$$F^q \notin J_{\le 17}.$$

Thus these low-degree equations do not even define the full variety set-theoretically.

Proof. Define a point P by $X_r(P) = 1$ for the 11 row types occurring in A, and $X_r(P) = 0$ for all other flows. This point has all variables outside E zero. Proposition 4 shows that every element of $J_{\le 17}$ vanishes at P, but $F(P) = 1 - 0 = 1$. No positive power of F can therefore lie in $J_{\le 17}$.

This is a boundary point, with many coordinates zero. No claim is made about the analogous question after restricting to the locus where all coordinates are nonzero. Also, no equality $\phi(\mathbb{Z}/17\mathbb{Z}) = 18$ is asserted; the lower bound is sufficient to disprove the proposed inequality.

Reproducible verification

The accompanying file verify_phylogenetic_Z17.py uses only the Python standard library. It checks all 173 triples to construct the allowed row types, the row sums and column multiplicities, an exact rational row reduction, and a full enumeration of compatible count vectors. It returns exactly the two vectors α, β.

An additional exact determinant certificate is available. In the matrix M, delete rows 16 and 23 and column 22, numbering from 1 with the ordering specified above. The resulting 21×21 determinant is 17. Together with $M(\beta - \alpha) = 0$, this certifies rank 21. The elementary proof does not depend on trusting this numerical determinant claim.

The separately written audit_Z17_independent.py uses a different check. It fixes the 18 first-column entries in increasing order and counts all second- and third-column fillings by dynamic programming on remaining multiplicities. There are 96 labelled fillings in total. Table A accounts for 24 and table B for 72, by multinomial counting within equal first-entry blocks. Every compatible unordered table has at least one such filling, so no third table exists. This check uses neither matrix rank nor the one-dimensional-kernel conclusion; it visits 981 memoized states. All counts are exact integers.

Verification status

The displayed argument is a complete mathematical proof, with finite arithmetic certificates and independently implemented checks. The note was prepared with AI assistance. Independent expert review and priority assessment have not yet been obtained. The material has not been submitted or published by the assistant.

References

[1] Mateusz Michalek and Emanuele Ventura, Finite phylogenetic complexity and combinatorics of tables, Algebra & Number Theory 11 (2017), 235-252. arXiv:1606.07263.

[2] Bernd Sturmfels and Seth Sullivant, Toric ideals of phylogenetic invariants, Journal of Computational Biology 12 (2005), 204-228. arXiv:q-bio/0402015. Conjecture numbering in this note refers explicitly to the arXiv version.